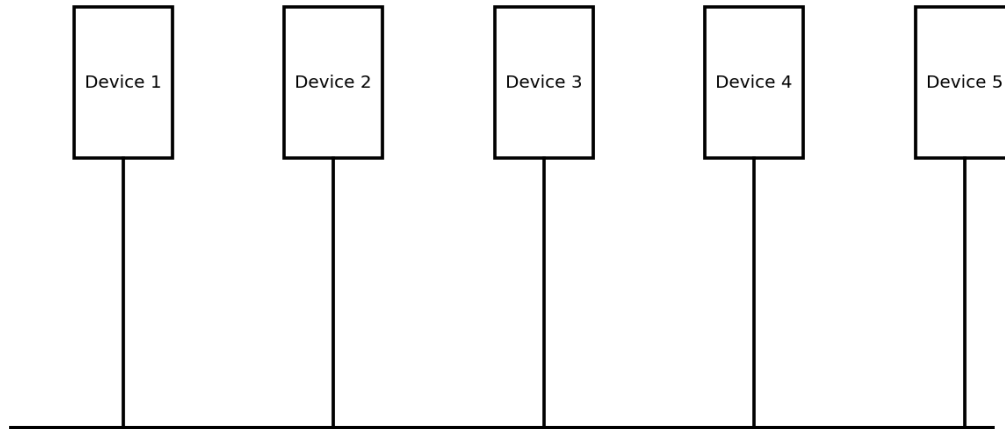


1) Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

Bus Topology Diagram

Bus Topology



Main Bus / Backbone Cable

Key Components

- Main bus/backbone cable — the shared communication path.
- Devices/nodes — devices connected to the main bus.
- Connection points/taps — points where devices connect to the shared cable.

Real-World Example

- A traditional Ethernet network using a shared coaxial backbone is an example of a bus-style network. This topology was used in older small LAN installations.

2) In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

Step 1: Open Cisco Packet Tracer

- Open Cisco Packet Tracer and create a new project.
- Place one switch in the workspace.
- Place four PCs around the switch.

Step 2: Connect the PCs to the Switch

- Use Copper Straight-Through cables.
- Connect each PC FastEthernet0 interface to a separate switch port.
- Arrange the devices around the switch to form a star-shaped layout.

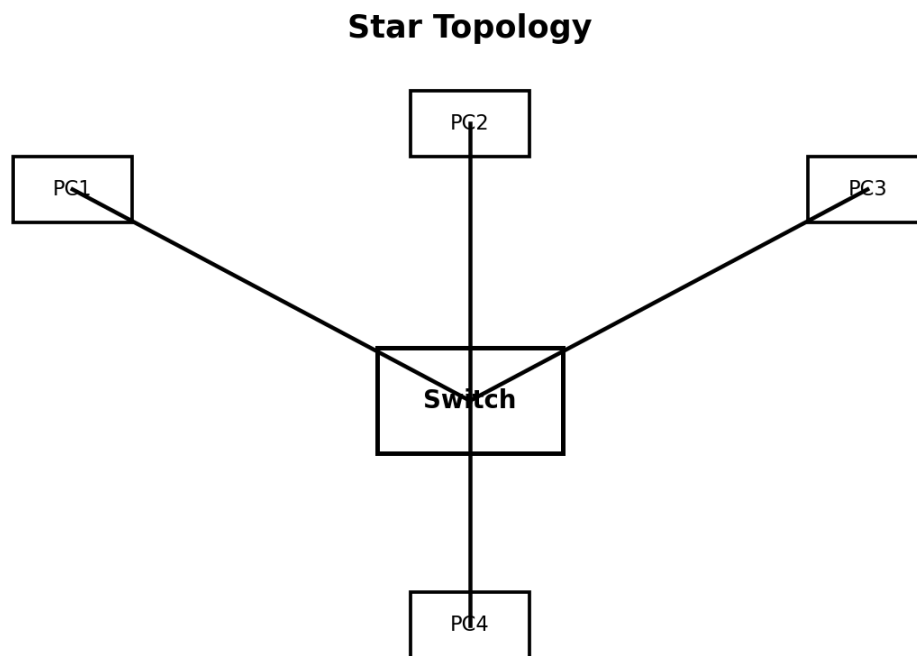
Step 3: Rename the PCs

- PC0 → WhatsApp
- PC1 → Zomato
- PC2 → Spotify
- PC3 → Flipkart

Step 4: Verify the Topology

- Confirm that all four PCs have an active connection to the central switch.
- The switch acts as the central connecting device in the star topology.

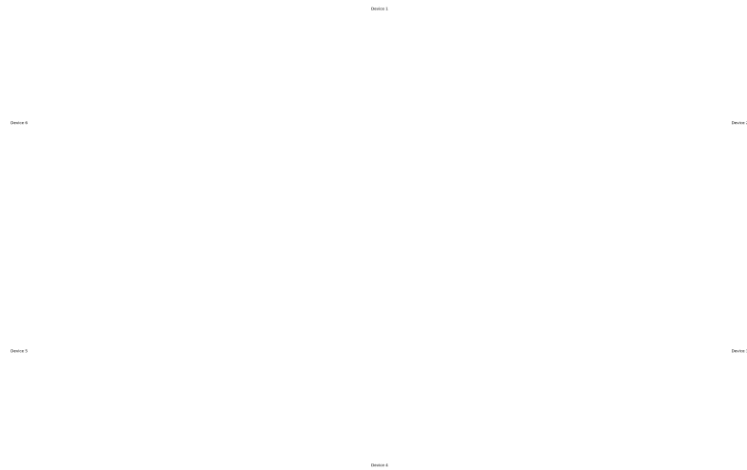
Reference Star Diagram



3) Draw a ring topology diagram and explain in 2–3 sentences how data travels between devices in this setup.

Ring Topology Diagram

Ring Topology - Data Flow Direction



How Data Travels

- In a ring topology, each device is connected to two neighboring devices, forming a closed loop.
- Data travels from one device to the next around the ring until it reaches the destination device; the arrows in the diagram show one possible direction of data flow.
- If the network uses a bidirectional ring, data may also travel in the opposite direction depending on the implementation.

4). Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.

Comparison Table

Topology	Advantages	Disadvantages	Typical Use
Mesh	1. High reliability and redundancy. 2. Multiple paths can keep communication available if one link fails.	1. Requires more cables and ports. 2. More expensive and complex to install.	Critical networks where reliability is important.
Star	1. Easy to install and manage. 2. Failure of one device cable usually affects only that device.	1. Failure of the central switch can affect the whole network. 2. Requires a central device and separate links.	Modern LANs, offices, labs, and home networks.

Conclusion

- Bus, star, ring, and mesh are different network topology designs. The practical activity demonstrates how a star topology is built in Cisco Packet Tracer and how the physical/logical arrangement of devices affects network communication and reliability.